



Assessment of the quality, readability, and usefulness of ChatGPT generated medical information for ten common cancer types

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Abstract

Chat Generative Pre-trained Transformer (Chat GPT) is an open platform to access different types of information in various areas including medicine. In this study we aimed to assess the quality, readability and usefulness of ChatGPT - generated medical information for the ten common cancer types. Seven questions for ten common types of cancer were asked to Chat GPT, and the medical information provided was evaluated with the Global Quality Scale (GQS). The information provided is classified as “useful” or “misleading”. The readability of the answers generated by Chat GPT was assessed using the Flesch–Kincaid Grade Level (FKGL) test and the Gunning Frequency of Gobbledygook (Gunning Fog) formula. Among all 70 answers the mean scores were 4.03 ± 0.48 for GQS, 15.11 ± 7.36 for FKGL and 18.16 ± 7.40 for Gunning Fog. Across all questions 4.3% of the total content found as misleading. Answers for questions about colorectal cancer exhibited the highest readability score (median FKGL = 21.5, min = 10.0–max = 25.7, median Gunning Fog = 23.7, min = 10.8–max = 29.4). Answers for skin cancer questions had the lowest readability (median FKGL = 11.4, min = 8.1–max = 46.9, median Gunning Fog = 14.2, min = 11.2–max = 49.8). The readability was assessed as “difficult”. The ChatGPT platform can represent a source of information as it provides highly accurate data. The ChatGPT platform can represent a source of information as it provides highly accurate data. However, readability was “difficult”. Although we think that the platform may be useful for medical students and others in the medical field seeking health information, it may not be suitable for the public for the same purpose.

Keywords ChatGPT · Quality · Cancer · Artificial intelligence · Information

1 Introduction

Artificial intelligence (AI) aims to endow machines with the ability to mimic human intelligence [13]. This technology consists of computer software that uses a complex

mathematical algorithm to process input data and produce predefined outputs yielding relevant results [26]. AI supports humans in terms of decision-making, needs assessment, planning, resource allocation, and monitoring large quantities of highly complex data [33].

As in many other sectors, the use of artificial intelligence software in decision-making processes in the fields of health and education has become a topic of discussion. Additionally, it can be used to identify gaps to assist children with intellectual disabilities in terms of those areas [1, 11]. Language models have been investigated in the medical sphere as instruments for individualized patient engagement and consumer health education [6, 28]. In November 2022, the AI software company OpenAI released a new open-source, the natural language processing tool known as Chat Generative Pre-trained Transformer, or ChatGPT [9]. Only a few days after being launched, ChatGPT had already attracted more than one million users [15].

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ChatGPT is a 175 billion-parameter natural language processing (NLP) model that produces user-prompted responses using deep learning algorithms that have been trained on significant quantities of data. As a dialogue agent, ChatGPT is designed to respond to a wide range of topics [10]. With the addition of supervised and reinforcement learning methods, ChatGPT expands on OpenAI's earlier GPT-3.5 language models [29]. ChatGPT is the first in a new series of models combining clinical knowledge and dialogic interaction, resulting in a significant advance in the field of NLP [10, 29]. Individuals currently spend a substantial proportion of their time interacting with or using the internet as a source of health information and numerous forms of media [34]. Innovative web-based applications and developments in online intelligent devices and social media platforms are also helping experts in the fields of disease surveillance and outbreak detection, behavior monitoring, and public health communication and education [30].

Again, according to a study conducted in the USA, it was found that 36.3% had a little - missing trust in internet information about cancers [3]. In the study, which looked at the information analysis on German websites used by cancer patients as a resource, it was concluded that the fact that the internet has become an important source of information about cancer treatments for patients may lead to misinformation and wrong decisions [25]. In a meta-analysis, despite the perceived positive aspects of online health information, 31% felt that it was overwhelming, 76% felt that it displayed conflicting medical information about their cancer, and 27% felt that it confused them about the right cancer treatment [8].

According to the World Health Organisation, the common cancer types are breast, lung, prostate, colorectal, bladder, skin, endometrial, cervical, pancreatic, and gastric cancer [36].

In general, these 10 cancers affect all segments of society. It is not uncommon to use the internet to search for information about cancer. It has been reported that 80% of US adults reportedly using the internet to seek health information [2].

The purpose of this study was to evaluate the quality and readability of information on the ChatGPT platform, which can be used as a resource for individuals seeking health- and disease-related information about cancer. The results may also yield useful insights for auditors to prevent the dissemination of false information.

2 Material and methods

In this content analysis study, the answers listed on the ChatGPT platform (<https://openai.com/blog/chatgpt/>) on 01 February 2023 were analyzed. Seven different questions (definition, causes, diagnosis, symptoms, prevention, treatment and risk factors) had been generated for ten different types of cancer (breast, lung, prostate, colorectal, bladder, skin, endometrial, cervical, pancreatic, and gastric) and data was entered to the ChatGPT engine by two physicians (BMM and KE). Seventy answers in total were recorded in a file for subsequent evaluation. The content of the answers was analyzed for the quality, usefulness and readability by two expert independent reviewers (BMM and KE). Inter-rater reliability was evaluated by Kappa analysis. The data was presented as mean, standard deviation, median and minimum–maximum values.

The following questions were put to the platform for each of the ten cancer types mentioned above.

Question 1-What is (type of cancer)?

Question 2-What is the cause of (type of cancer)?

Question 3-How is (type of cancer) diagnosed?

Question 4-What are the symptoms of (type of cancer)?

Question 5-How can (type of cancer) be prevented?

Question 6-What is the treatment for (type of cancer)?

Question 7-What are the risk factors for (type of cancer)?

2.1 Quality

The Global Quality Scale (GQS) designed by Bernard was used to assess the quality of the text content. This scale is scored from 1 to 5 points, with higher scores indicating higher content quality (Table 1) [30].

2.2 Usefulness

The answers given by two independent researchers in accordance with the purpose of the study were examined, analyzed in terms of content, and classified as useful or misleading as follows.

Useful - those containing scientifically completely accurate information about any aspect of the disease: definitions,

Table 1. Global Quality Scale

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- | |
|--|
| 1. Poor quality, poor flow, most information missing, not helpful for patients |
| 2. Generally poor, some information given but of limited use to patients |
| 3. Moderate quality, some important information is adequately discussed |
| 4. Good quality good flow, most relevant information is covered. useful for patients |
| 5. Excellent quality and excellent flow, very useful for patients |
-

causes, diagnosis, symptoms, prevention, treatment, risk factors.

Misleading - containing scientifically unproven or incomplete information (e.g., the availability of a cancer vaccine).

We used the kappa coefficient to assess the level of agreement between the two researchers [17, 23, 27].

2.3 Readability

The website (<https://www.webfx.com/tools/read-able/>) was used for readability. Two validated indices that estimate text reading levels were used to assess readability, which can be defined as an individual's level of comprehension of a particular text. High readability scores indicate low readability of the text content [12].

Flesch–Kincaid Grade Level (FKGL) and Gunning Frequency of Gobbledygook (Gunning Fog) were used as readability indices in this study. Each indicator performs calculations using different formulae depending on word and sentence structures. The FKGL index is proportional to the average number of words per sentence and syllables per word, while Gunning Fog is proportional to the mean number of words per sentence and the percentage of polysyllabic words within those sentences [20–22, 24]. For FKGL, readability was classified as "easy" (grade 6), "average" (grade 6–10), or "difficult" (grade >10), in agreement with previous studies [16, 31, 35].

3 Results

The Quality and readability indicators and the usefulness of the answers were evaluated for all 70 responses. Among all 70 answers the mean scores were 4.03 ± 0.48 for GQS, 15.11 ± 7.36 for FKGL and 18.16 ± 7.40 for Gunning Fog (Table 2). Responses containing incomplete or incorrect information were categorized as "misleading". All correct and complete answers were considered "useful"; Useful group FKGL = 15.27 ± 7.46 , misleading group FKGL = 11.50 ± 3.45 , thus there was no significant difference between the two groups ($p = 0.436$); Useful group Gunning Fog = 18.27 ± 7.50 , misleading group Gunning Fog = 15.60 ± 4.67 , thus there was no significant difference between the groups ($p = 0.660$). Three (4.3%) of the responses were misleading and the remaining 67 (95.7%) were useful. The misleading responses were related with the following questions:

- What is the prevention of skin cancer ?
- What is the prevention of pancreatic cancer?
- What is the prevention of gastric cancer? (Fig. 1)

The kappa values used to determine agreement between the observers were 0.86 for GQS, 0.82 for FKGL, 0.81 for Gunning Fog, and 0.72 for usefulness. A histogram plot for readability indicators and GQS scale results is shown in Fig. 2.

The readability and quality of the answers were compared between the groups according to the different cancer types, as shown in Table 3. Answers for questions about the colorectal cancer exhibited the highest readability score (median FKGL = 21.5, min = 10.0–max = 25.7, median Gunning Fog = 23.7, min = 10.8–max = 29.4). Answers for the skin cancer questions had the lowest readability (median FKGL = 11.4, min = 8.1–max = 46.9, median Gunning Fog = 14.2, min = 11.2–max = 49.8). The median GQS score for the answers to the questions for 10 different cancer types was 4 for overall.

When we compare the readability and quality of the answers for 7 different types of questions (definition, causes, diagnosis, symptoms, prevention, treatment and risk factors) the highest readability score was measured for the answers for the cancer symptoms (median FKGL = 20.25 min = 11.2–max = 46.9, median Gunning Fog = 22.45 min = 14.6–max = 49.8). The responses to the questions about definition of each cancer type exhibited the lowest readability scores (median FKGL = 10.45, min = 8.1–max = 14.6, median Gunning Fog = 12.40, min = 10.4–max = 16.8) (Table 4).

4 Discussion

As biotechnology evolves, AI engines are being developed to explore technical improvements aimed at solving complex problems in almost every field of life, including science and medicine [4]. Chat GPT is a tool that can assist medical professionals in various ways including data analysis, risk assessment, screening and diagnosis. It can also provide medical information both for healthcare providers and public. However, the information provided is not necessarily accurate and the text generated is not always understandable and coherent. Therefore, there are several studies conducted to assess Chat GPT generated information for reliability, accuracy and readability [18, 19]. In our study, we looked at the accuracy, readability, and quality of the answers provided by ChatGPT for ten common cancer types.

Johnson et al. evaluated Chat GPT answers for 284 medical questions which were generated by 33 physicians across 17 specialties for accuracy and completeness. They found the answers were mostly accurate and complete [14]. In another study, which analyzed Chat GPT answers and compared it with the National Cancer Institute's "Common Cancer Myths and Misconceptions" webpage, the authors found 11 of 13 Chat GPT answers were accurate (96.9%).

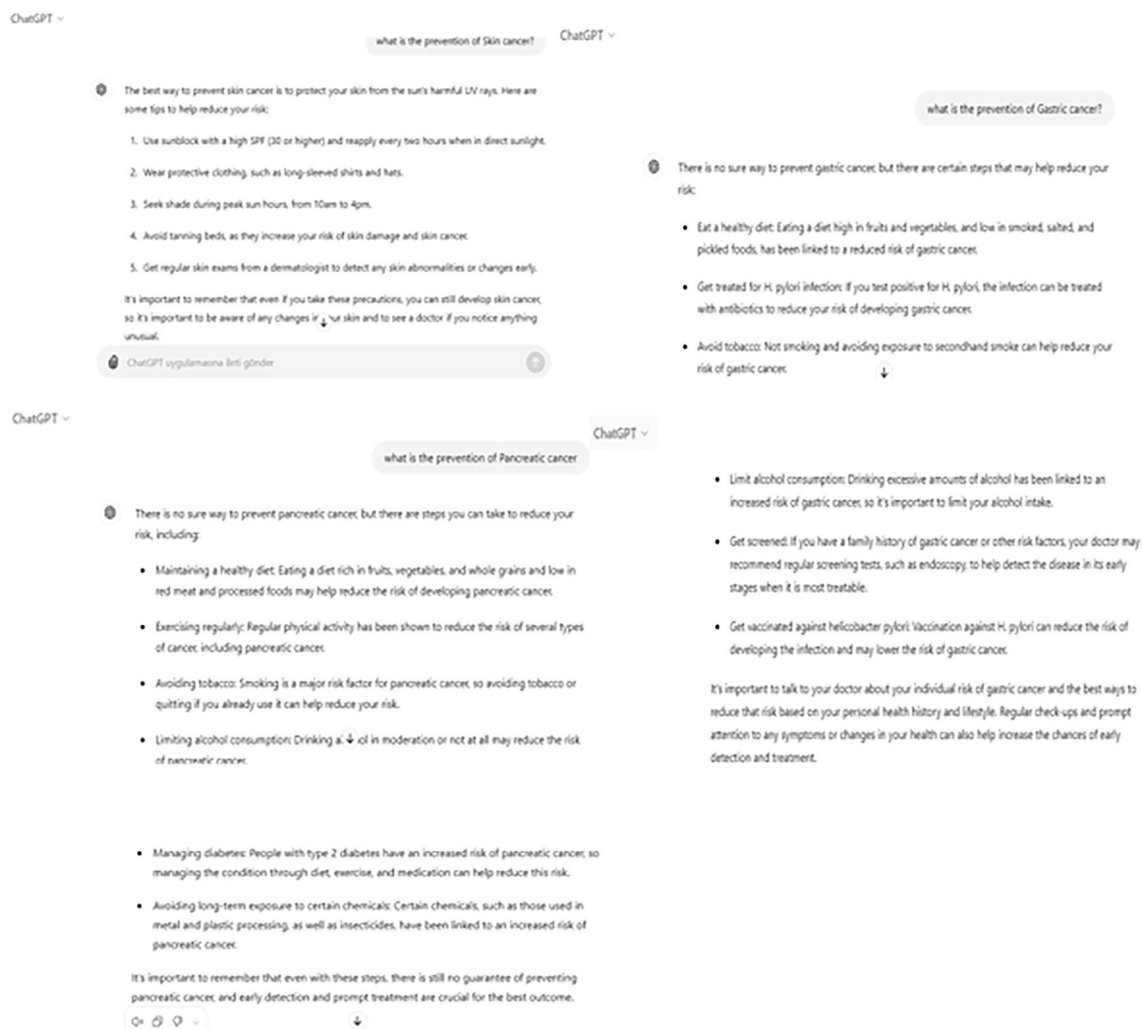


Fig 1. Questions and misleading answers

Table 2. Display of overall parameters

	Mean $\pm$ SD	Median (min–max)
FKGL	15.11 $\pm$ 7.36	12.35 (7.5–46.9)
Gunning Fog	18.16 $\pm$ 7.40	15.55 (10.4–49.8)
GQS	4.03 $\pm$ 0.48	4 (3–5)

FKGL Flesch-Kincaid Grade Level, GQS Global Quality Scale

However, they found both Chat GPT and NCI answers were less readable than would likely be recommended by health literacy advocates [32]. In our study, we received correct answers for 67 out of 70 questions. Although the quality and accuracy of the information provided were high, readability was found to be "difficult". In addition, answers to questions about the definition of each type of cancer exhibited the lowest readability scores. These results indicated that even the most basic question about

cancer types would be difficult for the general public to understand.

Therefore, despite being convincing in nature, ChatGPT text output may also be potentially disruptive or less readable [15]. Also, in another study assessing prostate cancer information content on the YouTube media platform reported a rate of potentially misleading content of 77% [7]. According to the results of our study, although the answers are useful information (%95.3), it should not be ignored that 4.3% of the answers produced by the Chat GPT for 7 different questions for 10 different cancer types contain misleading information. In addition, prevention and treatment answers do not have high readability scores, which would make it very difficult for the public to use the correct information.

The GQS score for information quality content in a testicular cancer study was 2.61 [7].

Our search of the literature revealed no previous studies evaluating the quality of information on the ChatGPT

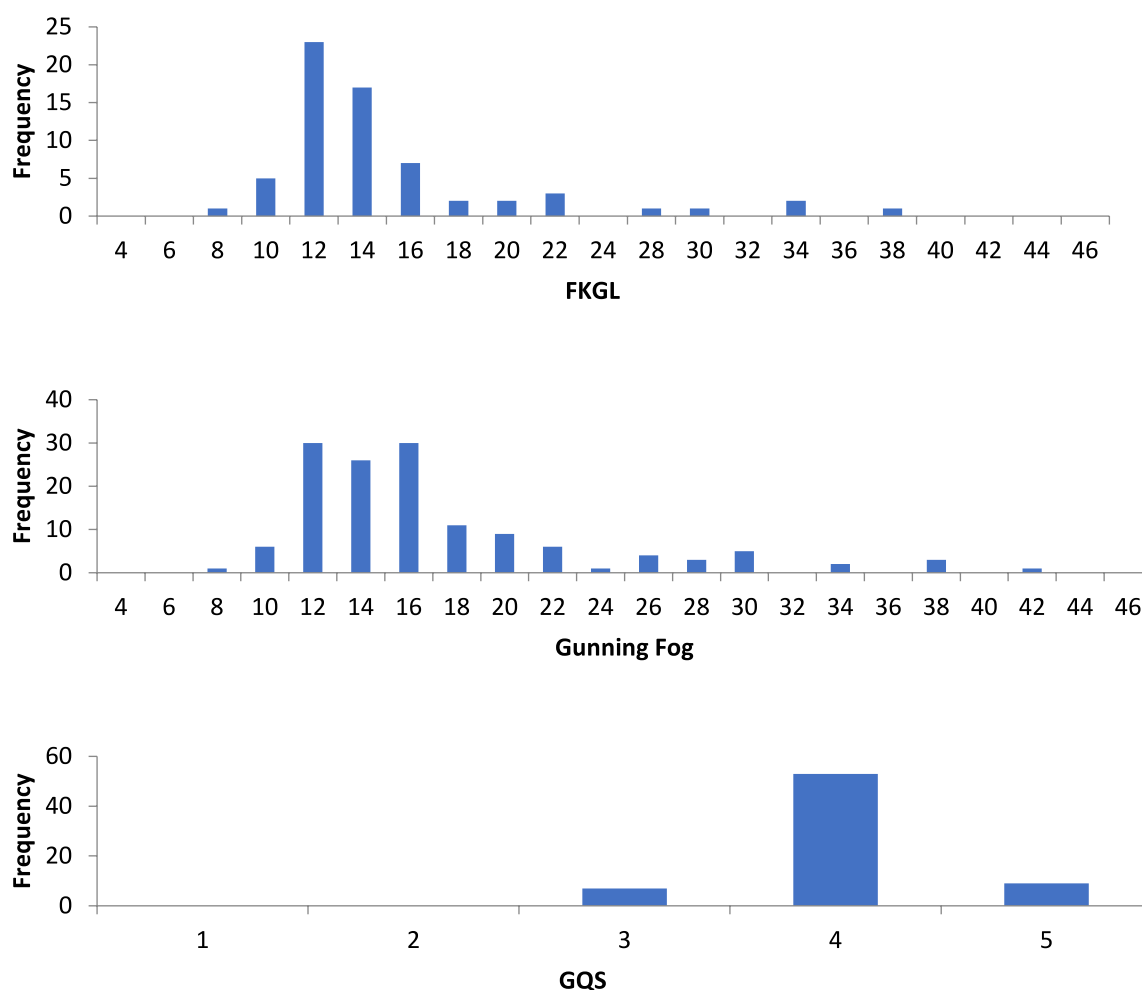


Fig 2. Readability indicators and GQS scale results

Table 3. Quality and readability parameters by cancer types

	FKGL	Gunning Fog	GQS
Breast cancer	12.4 (9.0–33.6)	16.8 (12.5–36.8)	4 (3–5)
Lung cancer	11.8 (8.1–22.0)	15.4 (10.4–29.1)	4 (3–5)
Prostat cancer	12.4 (7.5–18.4)	14.5 (11.0–20.1)	4 (3–5)
Colorectal cancer	21.5 (10.0–25.7)	23.7 (10.8–29.4)	4 (4–5)
Bladder cancer	12.3 (10.1–37.3)	14.9 (11.4–40.3)	4 (3–5)
Skin cancer	11.4 (8.1–46.9)	14.2 (11.2–49.8)	4 (3–4)
Endometrial cancer	13.0 (11.8–14.6)	15.6 (14.7–17.1)	4 (4–5)
Cervical cancer	12.1 (9.3–27.8)	15.6 (11.6–29.0)	4 (4–5)
Pancreatic cancer	11.9 (11.2–33.7)	16.3 (14.3–37.4)	4 (3–4)
Gastric cancer	11.7 (10.8–16.2)	15.1 (13.9–19.1)	4 (3–4)

FKGL Flesch-Kincaid Grade Level, GQS Global Quality Scale

platform with GQS. It may be assumed that the ChatGPT platform offers a large quantity of useful information with a high quality score. However, mechanisms for checking the quality and accuracy of online materials are often

Table 4. Quality and readability parameters for subtypes about cancer

	FKGL	Gunning Fog	GQS
Definitions	10.45 (8.1–14.6)	12.40 (10.4–16.8)	4 (3–4)
Causes	13.05 (11.2–24.7)	15.80 (14.0–29.1)	4 (3–5)
Diagnosis	13.55 (11.0–37.3)	17.15 (14.10–40.3)	4 (3–5)
Symptoms	20.25 (11.2–46.9)	22.45 (14.6–49.8)	4 (4–5)
Prevention	11.75 (7.5–21.5)	15.35 (11.0–23.7)	4 (3–5)
Treatment	11.85 (9.4–18.4)	15.75 (12.2–18.9)	4 (4–5)
Risk factors	12.05 (10.8–27.8)	15.20 (12.8–29.0)	4 (4–5)

FKGL Flesch-Kincaid Grade Level, GQS Global Quality Scale

either non-existent or very limited in scope. It is therefore important to be aware of the reliability of sources and the risk of misleading information being provided, and for this to be prevented [20–22]. Our study used the GQS scale for information quality content, and the score was 4.03. This result showed that the information was of high quality.

Individuals are currently employing all possible means to access information easily and quickly, and are extending the frontiers in the process [20–22]. We therefore consider it very important to use AI for easy access to information in the age of technology, the final, fully evolved form of which cannot be predicted. The fact that the level of misleading content is low and the information quality high within the scope of this can be considered an advantage of using AI in the health field. Focusing on the results of our research, we found that the level of misleading content was low and the quality of information was high. It supports the literature context.

Previous studies in which health-related web materials were analyzed have shown that information must be understandable in addition to accurate and high-quality [5, 16]. It is important that readers should be able to understand the information provided. Individuals of different socio-economic or cultural levels may interpret or understand what they read differently. We used two various readability indicators in the present study, and determined an average FKGL score of 15.11 and Gunning Fog score of 18.16. The readability assessment of the ChatGPT content analyzed in this study was "difficult". Content that is difficult to read will limit users' understanding, even if the information is of a high quality or accurate. It will therefore be easier for health professionals to use this platform for health information. A previous study also showed that ChatGPT possesses sufficient medical knowledge to allow it to be used as a new educational tool, one which may be especially useful for medical students [10].

The current study has a number of limitations. First, ChatGPT may provide different answers to the same questions at different times. We only asked questions to Chat GPT once and assessed the answers. We did not check the reproducibility of the answers or the effect of the repeated questions and time on answer quality or readability. Secondly, the dataset was limited to 70 questions, all related to cancer, a larger number of questions for different medical specialities might have led to different results. Additionally, observers' judgments may be subjective even though they are based on objective criteria.

5 Conclusions

In conclusion, the information quality of the ChatGPT platform was high, and the level of misleading content was very low. We believe that the low readability of the most basic questions (definition, prevention, and treatment) compared to other categories may be misleading or confusing for the public. However, readability was "difficult". Although we think that the platform may be useful for medical students and others in the medical field seeking

health information, it may not be suitable for the public for the same purpose. More detailed studies should be conducted on the subject with a wider range of question types.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10209-024-01155-6>.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interest The authors have no relevant financial or non-financial interests to disclose.

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